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Description - This section applies only to 1.5T or 1.0T systems installed with the Erbtec RF/Pen cabinet.

1- TOOLS REQUIRED

#2 Phillips nonferrous screwdriver (for all magnet enclosure power supply components)

#1 Phillips nonferrous screwdriver (for the high current power supply board)

1/4-inch standard nonferrous screwdriver

WARNING!

POSSIBLE PERSONAL INJURY! REMOVE POWER FROM THE RF/PEN CABINET BEFORE REMOVING ANY COMPONENTS FROM THE MAGNET ENCLOSURE POWER SUPPLY. VERIFY THAT ALL BREAKERS IN THE REAR OF CABINET ARE OFF, AND DISCONNECT MAIN POWER FEED CORD BEFORE OPENING CHASSIS FOR SERVICE. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE OCCUR. LOCK OUT AND TAG OUT THE RF/PEN CABINET.

2- SLIDING OUT MEPS

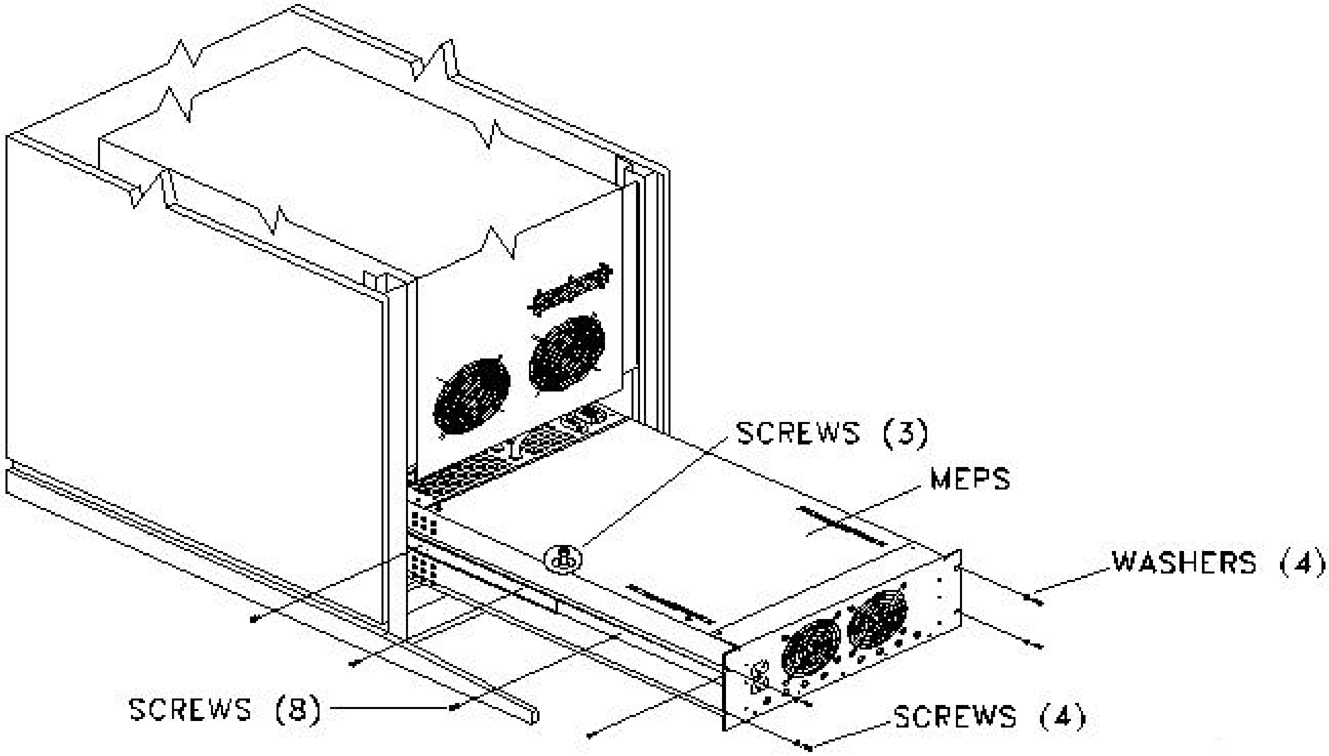
You will need to slide out the magnet enclosure power supply (MEPS) module in order to remove and replace any of the MEPS components (except the MEPS interface board). The MEPS interface board is located on the outside back of the module, and it is removed and replaced from the rear of the RF/Pen cabinet.

1. Turn off RF/Pen cabinet circuit breakers on rear of RF/Pen cabinet. Lock out and tag out the RF/Pen cabinet. (Refer to *Procedure For Safety: Section 6.*)

2. Verify that power to the RF/Pen Cabinet is off by checking the fans and LEDs on both the RF amplifier and the RF System Controller (RFSC).

3. Remove the front door from the RF/Pen cabinet.

4. Remove the four 10-32x1/2 panhead screws and four #10 flat washers securing the MEPS chassis to the RF/Pen cabinet. See Illustration L3108a.

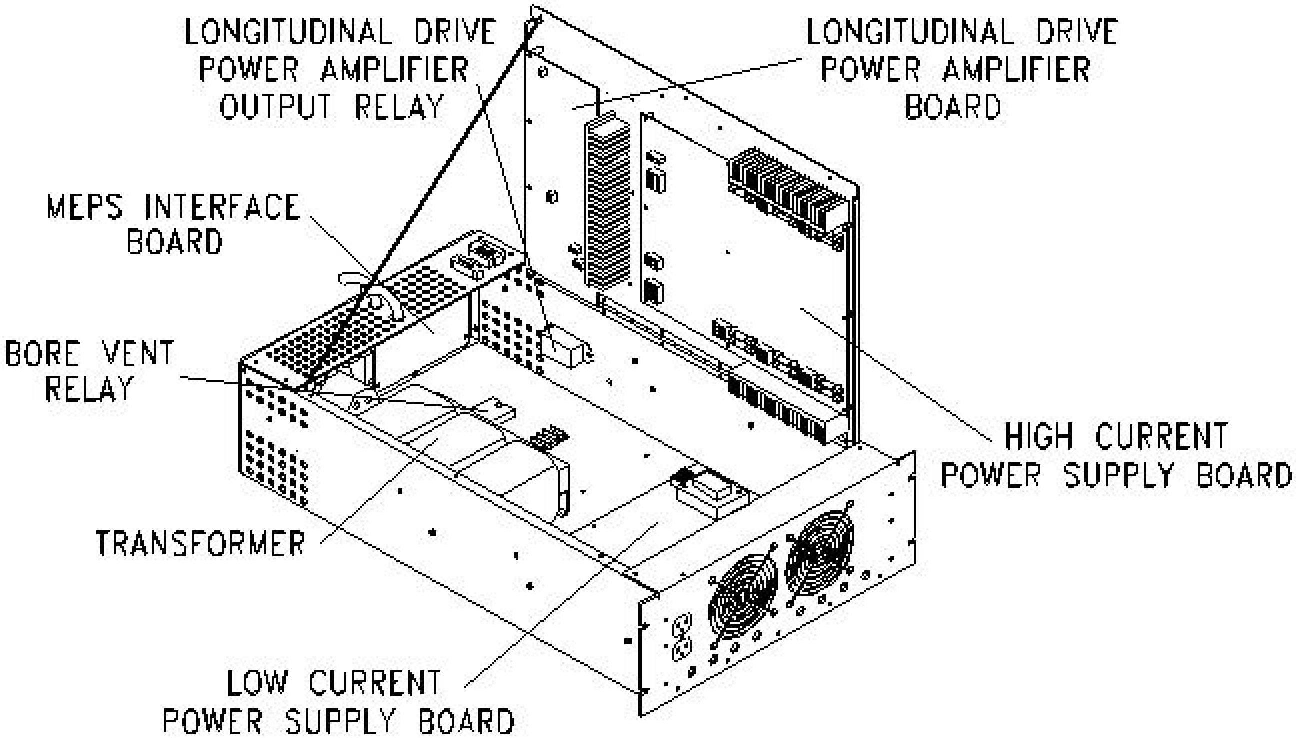


LOWER RF/PEN CABINET SHOWING MAGNET ENCLOSURE POWER SUPPLY
ILLUSTRATION L3108A

5. Slide out the MEPS from the RF/Pen cabinet (unless replacing the MEPS Interface Board).

6. Loosen the three 6-32 captive screws on the top cover of the MEPS module; see Illustration L3108a above.

7. Grasp the edge of the module lid on the left hand side and lift the lid to a vertical position. Illustration L3144a shows the location and names of the MEPS module components.



MAGNET ROOM POWER SUPPLY MODULE COMPONENTS
ILLUSTRATION L3144A

Go to the appropriate section to continue the remove and replace procedure:

- Transformer: See sections 3, 4.
- High Current Power Supply Board: see sections 5, 6, 7.
- Low Current Power Supply Board: see sections 8, 9, 10.
- Longitudinal Drive Power Amplifier Board: see sections 11, 12, 13.
- MEPS Interface Board: see sections 14, 15, 16.
- MEPS Module: see sections 17, 18.

3- REMOVAL OF FAILED TRANSFORMER

1. Verify that the power to the RF/Pen cabinet is off.

2. Disconnect all internal connections to the MEPS Interface Board Assembly.

3. Close the lid to the MEPS module and slide the unit back into the RF/Pen cabinet. Secure the MEPS module to the cabinet using one 10-32x1/2 panhead screw and one #10 flat washer.

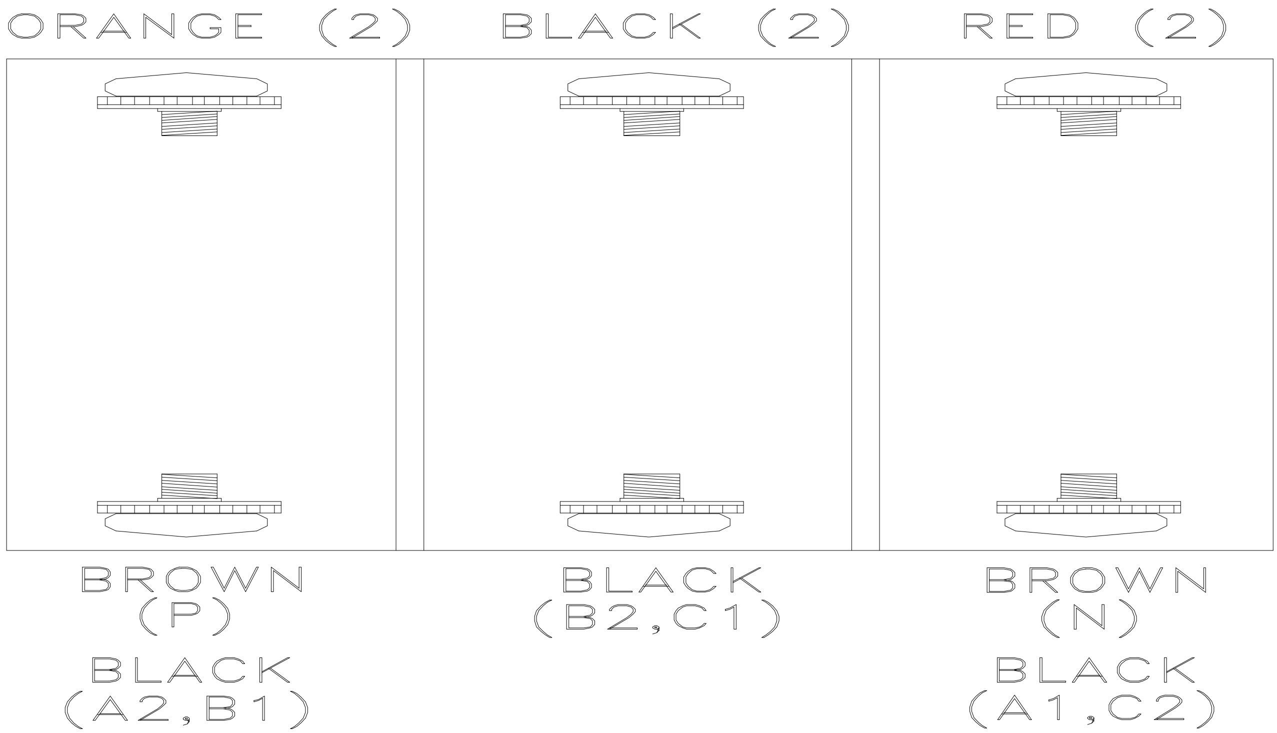
4. Disconnect all cables connected to the MEPS Module on the rear of the RF/Pen cabinet.

5. Remove the six 6-32x7/16 panhead screws attaching rear panel to the MEPS module.

6. Remove the one 10-32x1/2 panhead screw and one #10 flat washer from step 3.

7. Slide the MEPS module all the way out and lift the lid.

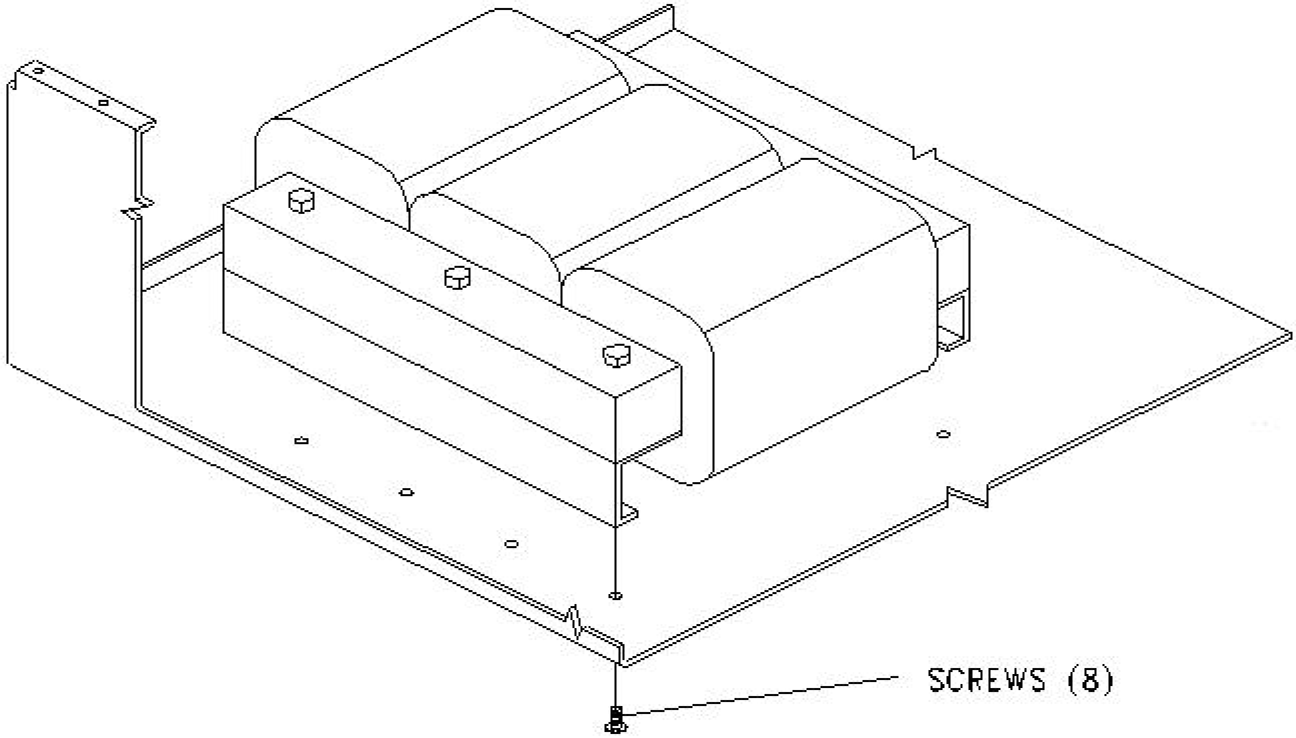
8. Turn the MEPS rear panel such that the bottom three slotted screws and star washers of CB6 can be removed. See Illustration L3179a.



CIRCUIT BREAKER #6
ILLUSTRATION L3179A

9. Remove ten 8-32x7/16 screws and cable clamps holding transformer harness.

10. Remove eight 10-32x3/8 screws and star washers securing the transformer to the bottom panel. See Illustration L3124a.



MAGNET ROOM POWER SUPPLY TRANSFORMER
ILLUSTRATION L3124A

11. Disconnect P1, P3, P5 and P6 from the High Current Power Supply Board.

12. Remove the failed transformer.

4- INSTALLATION OF NEW TRANSFORMER

1. With the MEPS module already out of the RF/Pen cabinet, and the lid open, position the new transformer inside the chassis.

2. Install eight 10-32x3/8 screws and star washers through the bottom panel of the MEPS Module. Tighten.

3. Connect P1, P3, P5, and P6 to the High Current Power Supply Board. Orientation is built into the harness.

4. Attach the ten cable clamps, securing the transformer harness with ten 8-32x7/16 screws.

5. Connect all internal connections to the MEPS Interface Board Assembly.

6. Turn the MEPS rear panel such that the bottom three slotted screws and star washers of CB6 can be attached (see Illustration L3179a see Section 3).

7. Close the lid and tighten the three captive screws to secure the lid.

8. Slide the MEPS Module all the way in. Be careful not to pinch any wires.

9. Secure the MEPS Module with one 10-32x1/2 panhead screw and one #10 flat washer.

10. Install six 6-32x7/16 panhead screws attaching the rear panel to the MEPS module.

11. Reconnect all cables connected to the MEPS module on the rear of the RF/Pen cabinet.

12. Secure the rest of the MEPS Module to the cabinet using three 10-32x1/2 panhead screws and three #10 flat washers.

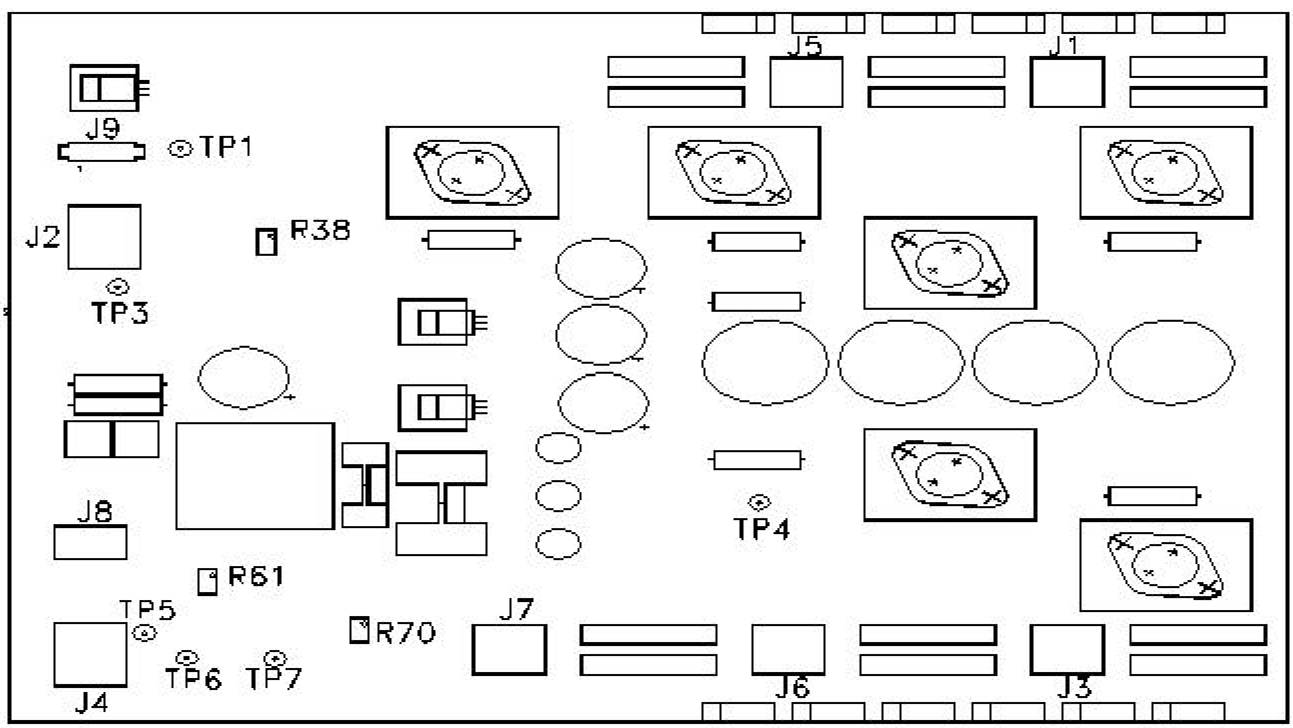
13. Replace the front door.

14. Remove lockout and tagout, and restore power to the RF/Pen cabinet.

15. Check functionality of the new transformer. Refer to procedure for MEPS Checks & Adjustments, Section 7, Patient Alignment Light, Bore Light , and Longitudinal/Dock Voltage Adjustments, Table 2.

5- HIGH CURRENT POWER SUPPLY BOARD SETUP

See Illustration L3149a



HIGH CURRENT POWER SUPPLY BOARD LAYOUT
ILLUSTRATION L3149A

See Table 1 & Table 2

TABLE 1
HIGH CURRENT PS TEST POINT INFORMATION

Test Point	Location Schematic	Name	Notes
TP 1		GND	
TP2		+15V	On Board Supply
TP3		+38.5V	
TP4		GND	
TP5		+15V Patient Alignment Light Output	
TP6		+12V Bore Light Output	
TP7		GND	

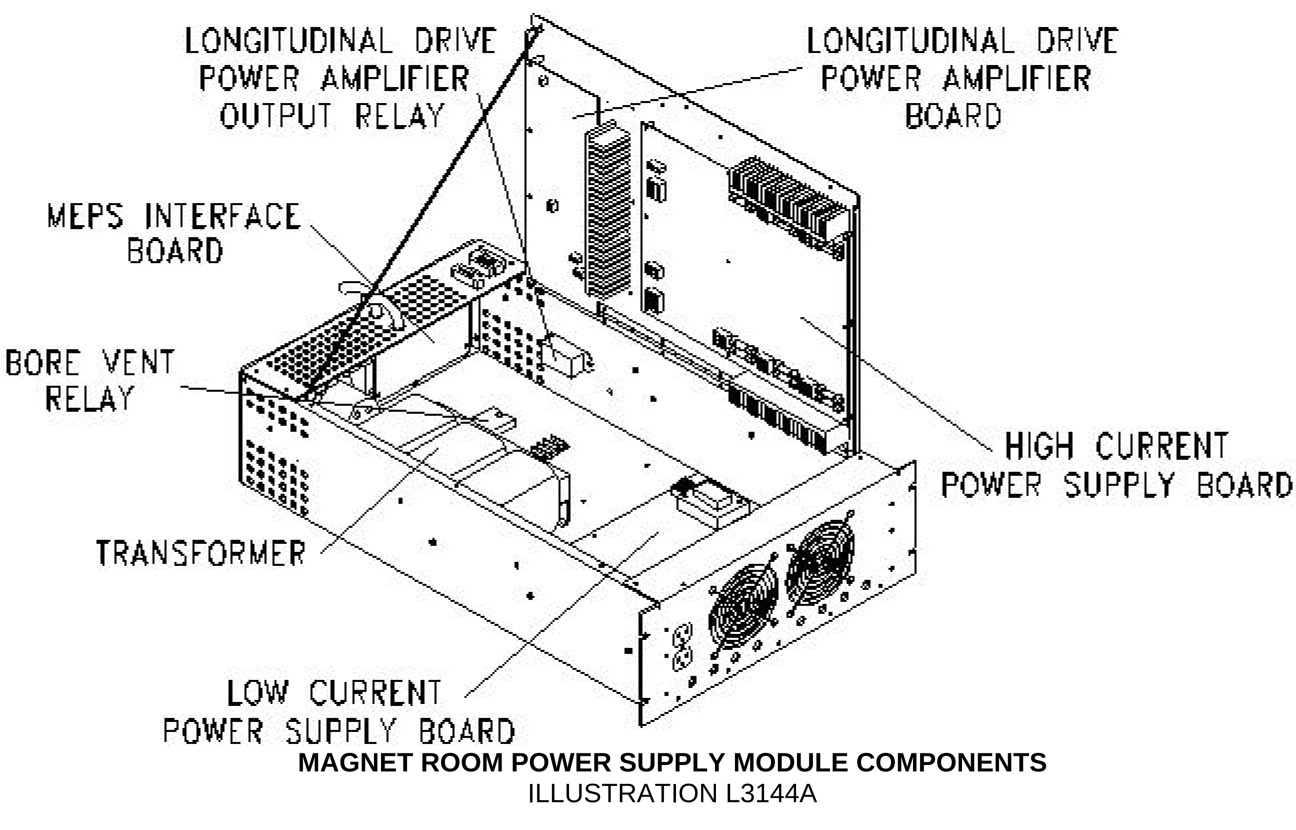
TABLE 2
HIGH CURRENT PS ADJUSTMENT INFORMATION

Test Point	Location Schematic	Name	Notes
R38		+38.5V Longitudinal/Dock	
R61		+12V Bore Light	
R70		+15V Patient Alignment Light	

6- REMOVAL OF FAILED HIGH CURRENT POWER SUPPLY BOARD

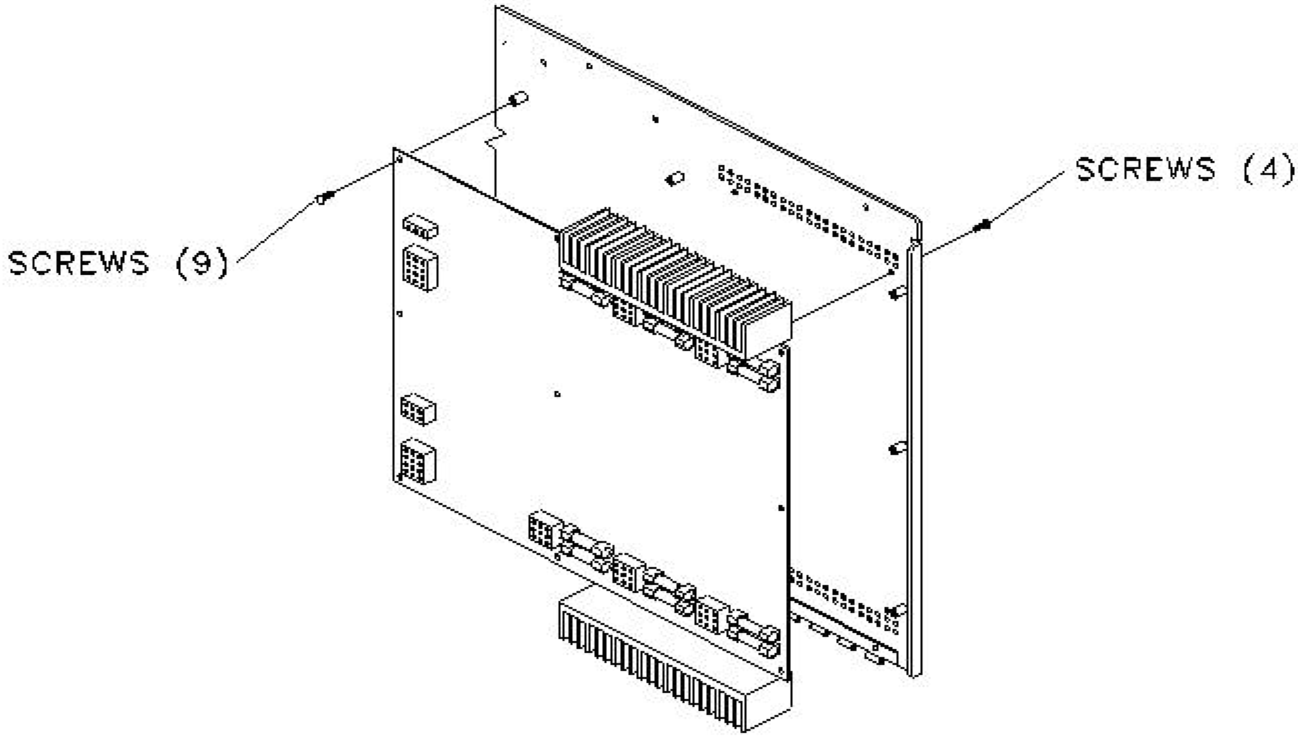
Verify that power is off and the RF/Pen cabinet is locked out and tagged out.

1. The High Current Power Supply Board is located on the top cover toward the front. See Illustration L3144a.



MAGNET ROOM POWER SUPPLY MODULE COMPONENTS
ILLUSTRATION L3144A

2. Disconnect all connectors attached to High Current Power Supply Board. See Illustration L3120a.



HIGH CURRENT POWER SUPPLY BOARD
ILLUSTRATION L3120A

3. Remove the nine 6-32x5/16 panhead screws and the four 6-32x5/16 screws securing the heatsinks to the chassis.

4. Remove the failed High Current Power Supply Board.

7- INSTALLATION OF NEW HIGH CURRENT POWER SUPPLY BOARD

1. Place the High Current Power Supply on the top cover. Position the board on the nine threaded standoffs with the heat sinks to the front. See Illustration L3120a in Section 6.

2. Install four 6-32x5/16 panhead screws through the outside of the lid into the heatsinks on the board. Do not tighten.

3. Attach nine 6-32x5/16 panhead screws on the threaded studs. Tighten all screws from steps 2 & 3.

4. Attach all connections. See Illustration L3120a in Section 6 showing correct placement of connectors.

5. Close the lid and tighten the three 6-32 captive screws.

6. Slide the module back into the RF/Pen cabinet and secure with four 10-32x1/2 panhead screws and four #10 flat washers.

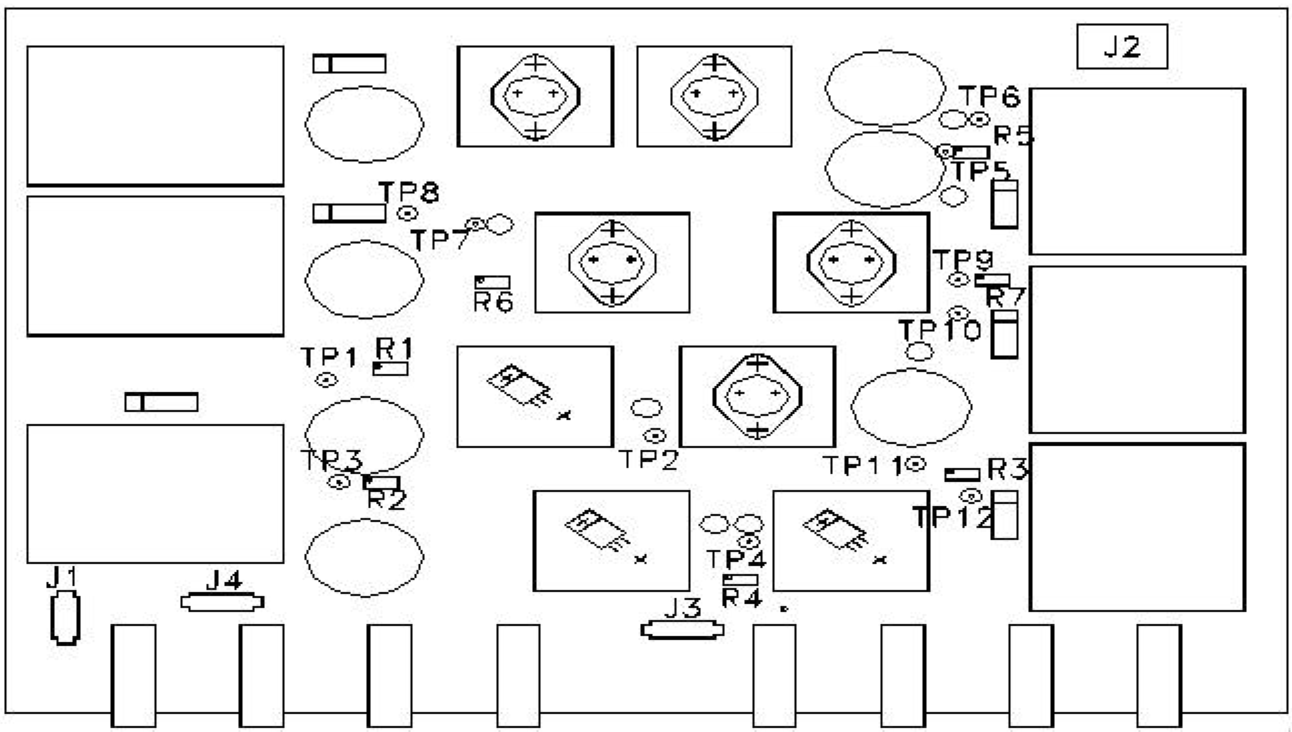
7. Replace front door.

8. Remove Lockout and Tagout and restore power to the RF/Pen cabinet.

9. Check functionality of the new High Current Power Supply board by referring to (Microsoft Word file RC1SCA4.DOC) procedure for MEPS Checks & Adjustments Table 2 - Patient Alignment Light, Bore Light and Longitudinal/Dock Voltage Adjustments.

8- LOW CURRENT POWER SUPPLY BOARD SETUP

See Illustration L3127a



LOW CURRENT POWER SUPPLY BOARD LAYOUT
ILLUSTRATION L3127A

See Table 3 & Table 4

TABLE 3
LOW CURRENT PS TEST POINT INFORMATION

Test Point	Location Schematic	Name	Notes
TP1		PAC +15V	
TP2		PAC CMN	
TP3		PAC -8V	
TP4		PAC -15V	
TP5		PAC +9V	
TP6		PAC +9V RTN	
TP7		SRI +24V	
TP8		SRI +24V RTN	
TP9		SRI +12V	
TP10		SRI +12V RTN	
TP11		SRI +8V	
TP12		SRI +8V RTN	

TABLE 4
LOW CURRENT PS ADJUSTMENT INFORMATION

Test Point	Location Schematic	Name	Notes
R1		PAC +15V	
R2		PAC -8V	
R3		SRI +8V	
R4		PAC -15V	
R5		PAC +9V	
R6		PAC +24V	
R7		SRI +12V	

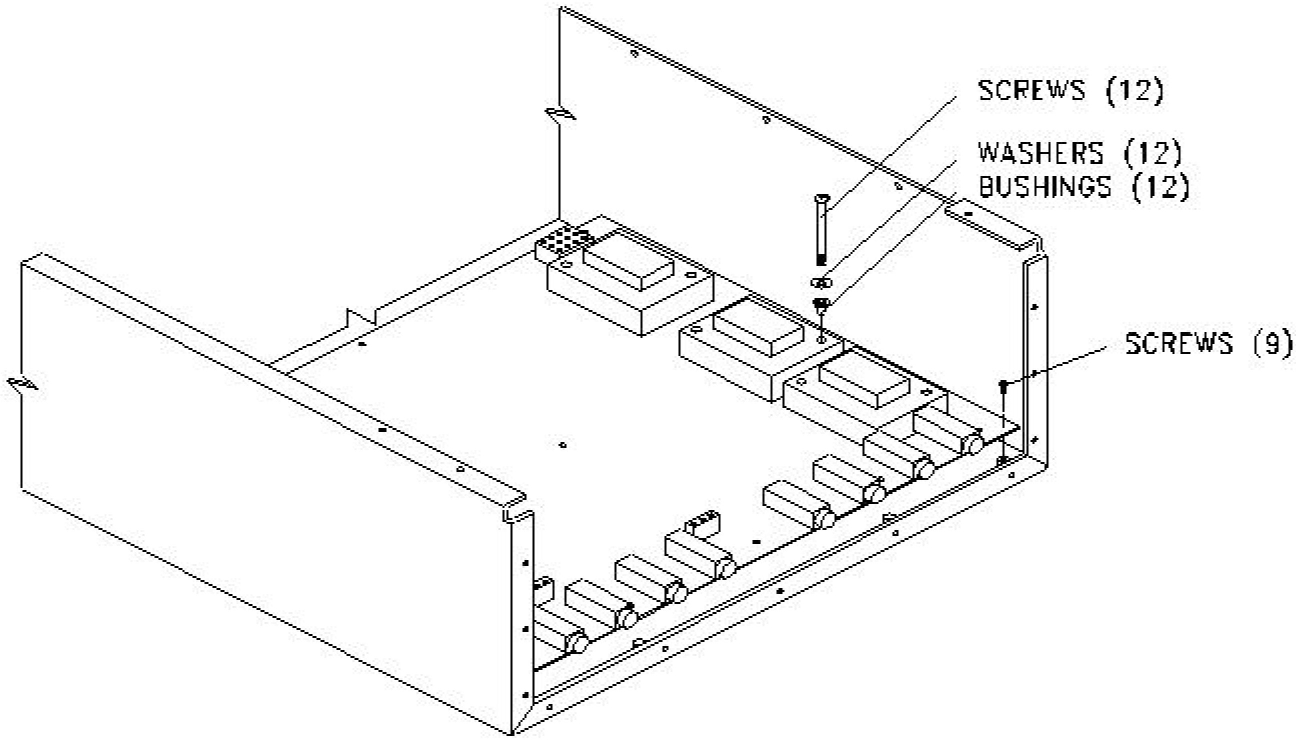
9- REMOVAL OF LOW CURRENT POWER SUPPLY BOARD

Verify that power is off and the RF/Pen cabinet is locked out and tagged out.

1. Disconnect all connectors attached to the Low Current Power Supply Board. See Illustration L3127a in Section 8.

2. Remove the thirteen 6-32 x 7/16 screws attaching the front panel and the one 6-32 x 3/8 screw on the front panel in the middle of the duplex connector.

3. Remove the nine 6-32x5/16 screws securing the board. See Illustration L3121a.



LOW CURRENT POWER SUPPLY BOARD
ILLUSTRATION L3121A

4. Remove the twelve 6-32x1-3/4 screws, #6 flat washers, and nylon bushings in the transformer holes.

5. Remove the Low Current Power Supply Board.

10- INSTALLATION OF NEW LOW CURRENT POWER SUPPLY BOARD

1. Place the new Low Current Power Supply Board in the bottom of the chassis with fuse holders facing the front. See Illustration L3121a in Section 9.

2. Attach nine 6-32x5/16 screws through pcb mounting holes. Do not tighten.

3. Install the twelve 6-32x1-3/4 screws, #6 flat washers, and nylon bushings into the transformer holes. Tighten all screws from steps 2 & 3.

4. Attach front panel with thirteen 6-32 x 7/16 screws and one 6-32 x 3/8 screw on the duplex connector.

5. Attach all connections to the board. See Illustration L3127a in Section 8 for connection information.

6. Close the lid and tighten the three 6-32 captive screws.

7. Slide MEPS module back into RF/Pen cabinet and secure with the four 10-32x1/2 screws and four #10 flat washers.

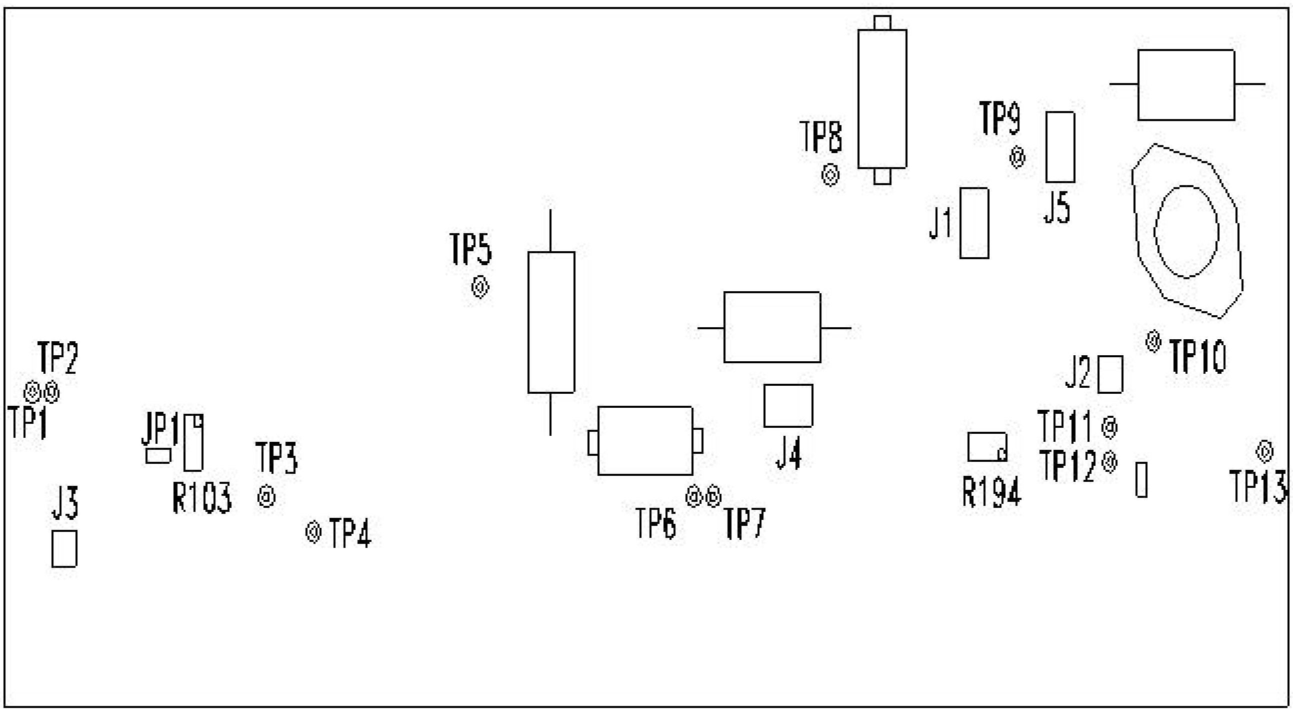
8. Remove lockout and tagout and restore power to RF/Pen cabinet.

9. Replace front door.

10. Check functionality of the new Low Current Power Supply board. Refer to Refer to procedure for MEPS Checks & Adjustments, Section 6, PAC and SRI Voltage Adjustments, Tables 3 and 4.

11- LONGITUDINAL DRIVE POWER AMPLIFIER BOARD SETUP

See Illustration L3173a.



LONGITUDINAL DRIVE POWER AMPLIFIER BOARD LAYOUT
ILLUSTRATION L3173A

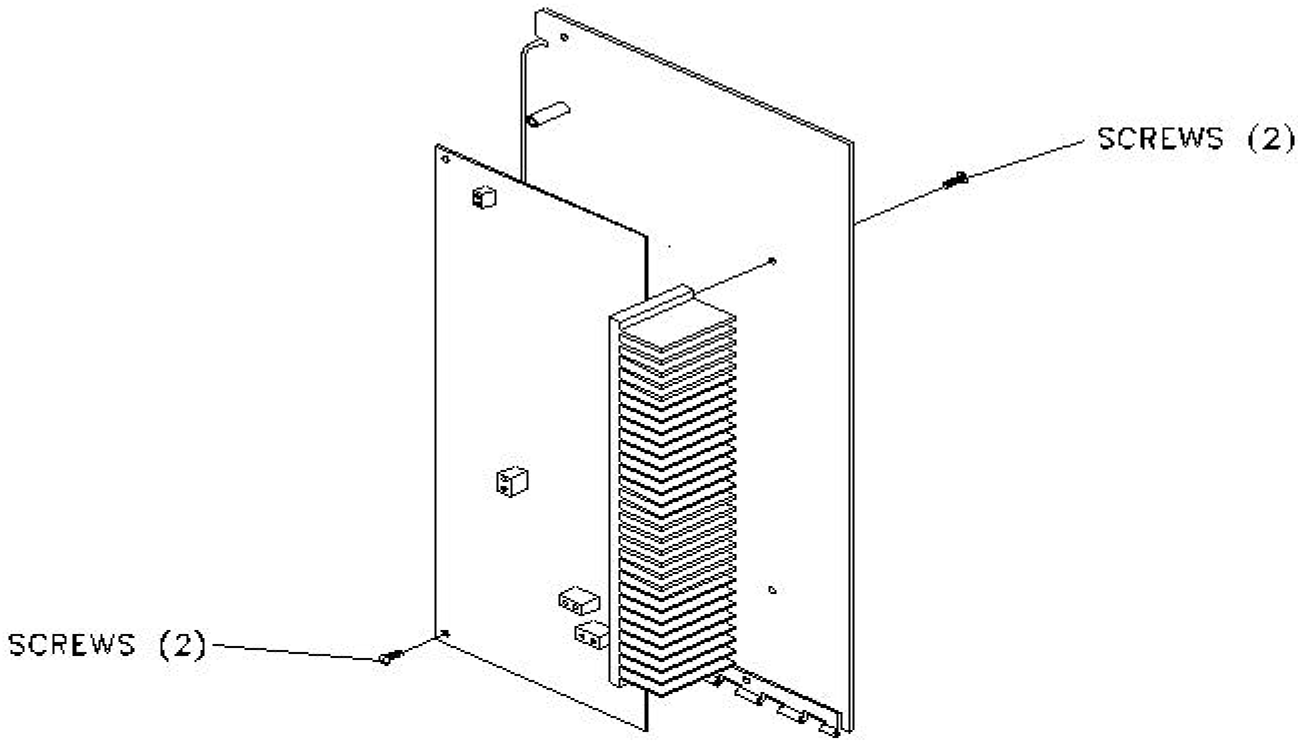
12- REMOVING FAILED LONGITUDINAL DRIVE POWER AMP

Verify that power is off, and that the RF/Pen cabinet is locked out and tagged out.

1. Disconnect all connectors attached to the Longitudinal Drive Amplifier Board . The Longitudinal Drive Power Amplifier Board is located at the rear of the top lid. See Illustration L3144a in Section 2.

2. Cut the wire tie securing the cable harness to the board.

3. Remove the two 6-32x5/16 screws securing the board to the standoffs and two 6-32x5/16 screws securing the heatsink. See Illustration L3122a.



LONGITUDINAL DRIVE POWER AMPLIFIER BOARD
ILLUSTRATION L3122

4. Remove the failed Longitudinal Drive Power Amplifier Board.

13- INSTALLATION OF NEW LONGITUDINAL DRIVE AMP BOARD

1. Install the new Longitudinal Drive Amplifier Board on the inside rear of the top lid. Secure with two 6-32x5/16 screws into the threaded standoffs. Do not tighten. See Illustration L3122a in Section 12.

2. Install the two 6-32x5/16 screws through the outside of the lid into the heatsink on the board. Tighten screws from steps 1 & 2.

3. Attach all connections. See Illustration L3173a in Section 11 showing correct placement of connectors.

4. Close the lid and tighten the three 6-32 captive screws.

5. Slide MEPS module back into RF/Pen cabinet and secure with four 10-32x1/2 screws and four #10 flat washers.

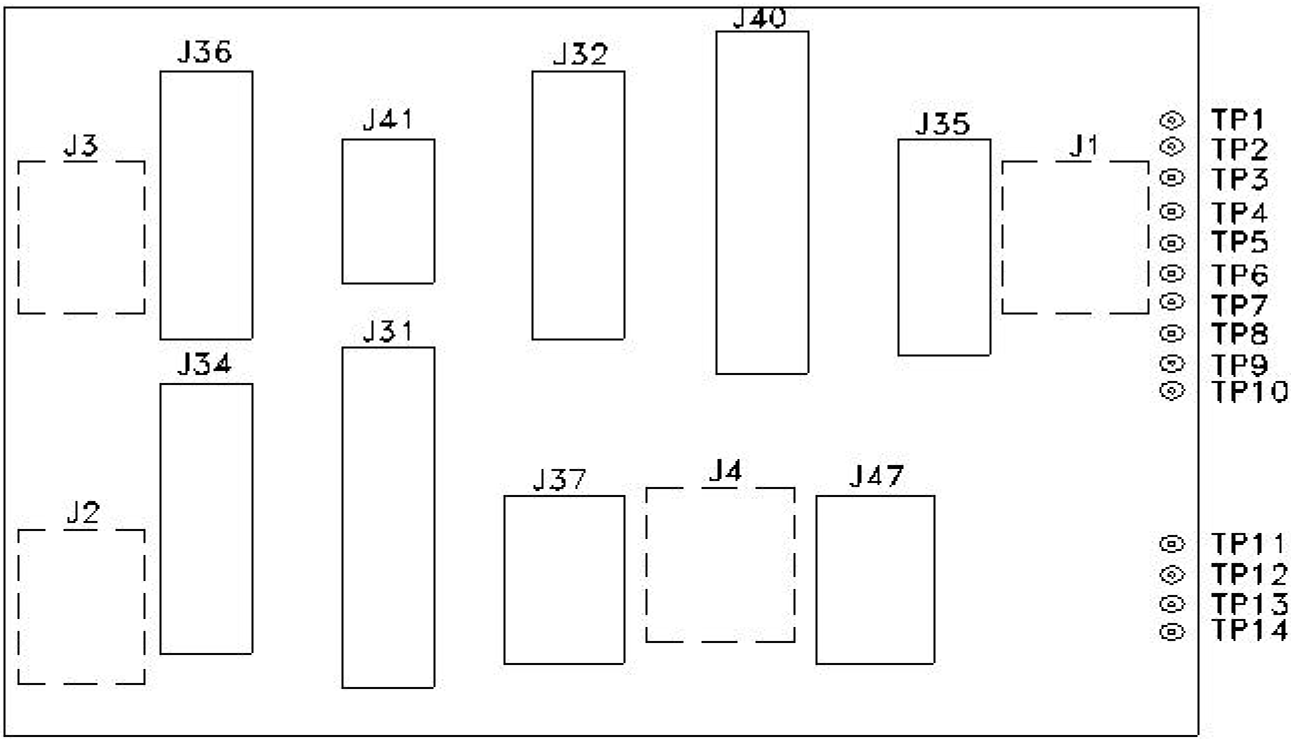
6. Remove Lockout and Tagout and restore power to the RF/Pen cabinet.

7. Replace front door.

8. Check functionality of the new Longitudinal Drive Amplifier Board by referring to the Procedure Longitudinal Drive System.

14- MEPS INTERFACE BOARD SETUP

See Illustration L3172a.

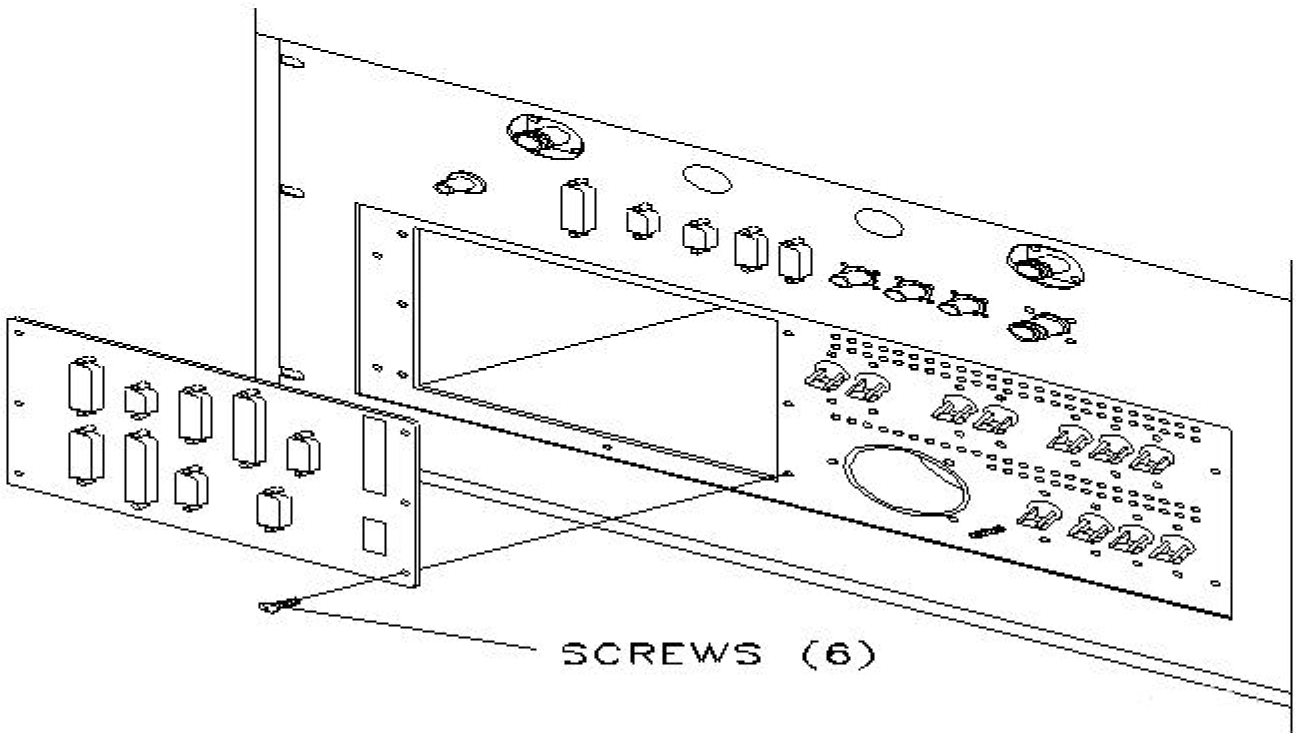


MEPS INTERFACE BOARD LAYOUT
ILLUSTRATION L3172A

15- REMOVAL OF FAILED MAGNETIC ENCLOSURE I/F BOARD ASSEMBLY

1. Verify that power is off and RF/PEN cabinet is locked out and tagged out.

2. Disconnect all external cables attached to the back of MEPS. Do not slide MEPS module out of cabinet. See Illustration L3123a.



MEPS INTERFACE BOARD
ILLUSTRATION L3123A

3. Working on the outside rear of the MEPS module, remove the six 6-32x3/8 screws located on the rear panel that are securing the Interface Board Assembly.

4. Pull Interface Board through the back of the MEPS module.

5. Disconnect all internal connectors.

16- INSTALLATION OF NEW MEPS I/F BOARD ASSEMBLY

1. Attach all internal connectors to MEPS interface board from the outside of the module. See Illustration L3123a see section 15.

2. Install the six 6-32x3/8 screws through the outside of rear panel. Tighten.

3. Connect all external cables to the MEPS interface panel.

17- REMOVAL OF MEPS MODULE

1. Turn off RF/Pen cabinet power circuit breakers on rear of RF/Pen cabinet. lockout and tagout the RF/Pen cabinet. (Refer to *Procedure For Safety: Section 6*.)

2. Verify that power to the RF/Pen Cabinet is off by checking the fans and LEDs on both the RF amplifier and the RF System Controller (RFSC).

3. Remove front door from RF/Pen cabinet.

4. Remove the four 10-32x1/2 screws and four #10 flat washers securing the MEPS chassis to RF/Pen cabinet.

5. Slide out MEPS module from RF/Pen cabinet.

6. Support MEPS module from underneath to alleviate pressure on the slides.

7. Remove the eight 8-32x3/8 screws (4 per slide) that attach the MEPS module to the slides. This will enable you to completely remove the unit. See Illustration L3108a in Section 2.

18- INSTALLATION OF NEW MEPS MODULE

1. Support MEPS module from beneath to alleviate pressure on slides.

2. Install eight 8-32x3/8 screws to secure the MEPS module to the slides of the RF/Pen cabinet.

3. Carefully slide the MEPS module back into the RF/Pen cabinet.

4. Install the four 10-32x1/2 screws and four #10 flat washers to the front of the MEPS module to secure the RF/Pen cabinet.

5. Install front door. Remove lockout and tagout and restore power to the RF/Pen cabinet.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 15, 1998	J. Saperstein	Initial conversion from Toolbook to Word.
1	May 21, 1999	SM Atladottir	Updated Procedure References for New GUI